NVIDIA Vera sandbox benchmarks

Where Vera stands against AMD EPYC 9575F, AMD EPYC 9J45 (Turin) and AWS Graviton5

Consolidated report · data as of 2026-10-15 11:15 UTC (adds Turin throughput and Graviton5 2,000-sandbox runs) · Daytona rl-platform benchmark cells

Scope. Daytona Linux sandboxes (Firecracker microVMs) on four bare-metal hosts running the **same Daytona build**, the same benchmark harness and the same recipes. Three measurements: **startup** (1,000 or 2,000 sandboxes created at once), **throughput** (jobs per second of a CPU-bound agent workload running inside packed sandboxes) and **density** (sandboxes alive at once). Plus a per-step breakdown of the create path, the fixes that changed the numbers, and the methodology corrections behind them. All four hosts are development benchmark cells, not production.

1. Executive summary

Vera leads every host on sustained CPU-bound sandbox throughput — including the dual-socket, 192-core AMD Turin — and has the fastest cores. It does not lead on sandbox startup, density or reliability.

Question	Leader	Evidence
Throughput (jobs/s)	**Vera**	Plateau: Vera 26.7 · Graviton5 22.9 · Turin 22.2 · EPYC 9575F 8.0. Same level, 2,000 sandboxes: Vera 26.1 · Graviton5 23.4 (Vera +12 %) · Turin 22.2 (+18 %) · 9575F 8.0. Vera beats dual-socket Turin with fewer cores (176 vs 192).
Speed per core	**Vera**	0.152 jobs/s per physical core vs 0.126 (9575F), 0.119 (Graviton5), 0.116 (Turin): +21–31 %. Fastest single job (5.9 s vs 6.6–7.1 s). Per hardware thread Graviton5 leads (0.119 vs 0.076).
Throughput per socket	Graviton5	Graviton5 22.9 on one socket vs Vera ≈ 13 and Turin ≈ 11 per socket (sockets differ in core count: 192 / 88 / 96).
Platform startup (idle image)	9575F, Graviton5	1k burst, typical sandbox, fixes on: Graviton5 0.85 s · 9575F 1.06 s · Vera 1.57 s · Turin 2.35 s.
Platform startup at 2,000	Graviton5	Typical sandbox: Graviton5 1.5 s · 9575F 2.4 s · Vera 2.9 s · Turin 3.5 s.
Startup under CPU load (agent image)	Turin	1k all ready: Turin 7.9 s · Graviton5 10–12 s · 9575F 19 s · Vera 22–28 s. 2k: Turin 21 s · Graviton5 27–29 s · Vera 50–55 s.
Density	Turin	8,000 live cleanly, peak 9,750 (limited by the runner heartbeat, not hardware). Vera and the 9575F verified to 2,000.
Reliability	Turin, Graviton5	Zero failures in every throughput ladder to 2,000 and in thousands of startups. Vera and the 9575F: 1 failure in 16,000 jobs at 2,000; Vera also ≈ 0.2 % of guests wedge after snapshot resume (3213).

Bottom line. The Turin ladder settled the open question in Vera's favour: on sustained throughput Vera is ahead of every host tested — +20 % over dual-socket Turin with 8 % fewer cores, +12–16 % over Graviton5, 3.3× the single-socket 9575F — and ahead on speed per core by 21–31 %. That is a clear, defensible lead on the work agents do inside sandboxes, though not an order-of-magnitude one. Vera does **not** lead on platform startup (Graviton5 and the 9575F are faster), startup under load or density (Turin), or failure rate (Turin and Graviton5 were cleaner). Open: jobs per watt and a memory-bandwidth-heavy workload (section 9).

2. Hosts and build

	HQ Vera	EPYC 9575F	Turin (OCI phoenix-e6)	Graviton5 (AWS m9gd)
CPU	NVIDIA Vera, 2 × 88 cores, SMT2 (352 threads)	AMD Zen 5, 1 × 64 cores, SMT2 (128)	AMD EPYC 9J45, 2 × 96 cores, SMT2 (384)	Neoverse-V3, 1 × 192 cores, no SMT
Memory	1,479 GiB	1,133 GiB	2,267 GiB	754 GiB
Host kernel / pages	6.17 nvidia / 64 KiB, `preempt=none`	7.0 / 4 KiB	6.17 oracle / 4 KiB	6.17 aws / 4 KiB
Guest kernel	6.12.34 arm64	6.8.12 x86	6.8.12 x86	6.12.34 arm64
Storage	1 NVMe	NVMe RAID	1 NVMe (RAID10 removed)	1 NVMe instance store
`RLP_VM_MAXACK`	256	**2,048**	256	256

Every host: Daytona `f0611fa13` + EROFS-4k (1060) + 30 s Firecracker socket wait (3212) + the four create-path fixes behind flags (section 6); Firecracker v1.16.1; daemon `rlp-rs-0.4.1`; balloon off; sandbox 1 vCPU, 0.025 CPU, 64 MiB; whole box, unpinned. Startup client: `mpfleet.py` with 16 processes. Results below are with the fixes **on** unless stated.

3. Results

3.1. Startup — 1,000 sandboxes at once

Metric (s)	HQ Vera	EPYC 9575F	Turin	Graviton5
Idle image: typical (p50), fixes off → on	2.26 → 1.57	2.51 → **1.06**	3.70 → 2.35	≈ 2.2 → **0.85**
Idle image: slow (p95), on	2.54–2.61	1.57–1.60	3.76	**1.54–1.56**
Idle image: all ready, on	2.7–3.9[a]	**2.05–2.26**	3.87–3.88	1.68–2.32[b]
Agent image: all ready, on	21.8–28.1[a]	18.8–19.1	**7.84–7.88**	10.1–12.3
Failures	1 in 3,000 (idle), 2 in 3,000 (agent)	0	0	0

[a] one run held open by a wedged guest (10.4 s idle, 61 s agent). [b] one run 7.6 s: 5 sandboxes with 5 s toolbox waits (new, fixes-on only).

3.2. Startup — 2,000 sandboxes at once

Metric (s)	HQ Vera	EPYC 9575F	Turin	Graviton5
Idle: p50 / p95	2.9 / 6.5	2.4 / 3.6	3.5 / 6.8	**1.5 / 3.0**
Idle: all ready	6.6–7.2 (one run lost)	4.6 (one run 17.7)	7.1–7.2	**4.4** (one run 15.4)
Agent: p50 / all ready	28.5 / 50–55	19.7 / 58–71	**7.0 / 21.4**	12.0 / 27–29
Failures	2 + 1 lost run	0	0	0

3.3. Throughput — dense agent ladder (jobs/s)

Published recipe: 45 jobs × 8 episodes per sandbox, hold-then-exec, 0.025 CPU (max 1), 64 MiB (max 4 GiB).

Host	44	176	352	704	1,056	2,000	failures
HQ Vera	—	15.2	**26.7**	**25.6**	—	**26.1**	1 at 2,000
Graviton5	5.7[c]	23.6	22.3	22.9	22.5	23.4	0
Turin	5.5	17.9	20.2	22.3	22.2	22.2	0 (11 levels)
EPYC 9575F	—	8.0	8.1	8.0	—	8.0	1 at 2,000
Vera one socket (published Sep)	6.0–6.6	11.8–13.7	12.9–13.3	12.8–13.1	12.8–13.0	12.7–14.2	53–627 at 2,000
EPYC 9755 (published)	4.3–5.5	9.6–10.0	10.7–11.5	10.7–11.3	10.7–11.3	10.9–12.0	from 1,408

All fixes on except [c] (fixes-off run; the fixes do not change throughput on any host). Turin ran all eleven levels (44 → 2,000) with zero failures; Graviton5 also ran all eleven (fixes off) with zero failures. Platform overhead is 0.5–0.6 % of each job, so throughput = cores × per-core speed under full load. At 2,000 a job takes 34 s (p50) on Vera, 80 s on Turin, 84 s on Graviton5, 239 s on the 9575F.

Derived	Plateau jobs/s	Cores / threads	Per core	Per thread	Single job, light load[e]	Clock drop packed[f]
HQ Vera	26.7	176 / 352	**0.152**	0.076	**5.9 s**	−6 %
Graviton5	22.9	192 / 192	0.119	**0.119**	6.65 s	−3 %
Turin	22.2	192 / 384	0.116	0.058	7.06 s	−21 %
EPYC 9575F	8.0	64 / 128	0.126	0.063	6.7 s	−24 %
EPYC 9755 (published)	≈ 11	128 / 256	0.086	0.043	7.1 s	—

[e] job p50 at 44 sandboxes. Vera's 5.9 s and the EPYC figures are from the published September runs (older build); Turin and Graviton5 from the current build. [f] not measured the same way: Turin pinned busy core at light load vs all-core mean packed; Graviton5 guest PMU counters; Vera from the OCI Vera probe (preliminary); 9575F perf counters. Treat as indicative. **Why Turin trails Vera:** it is slower per job even at light load (7.06 vs 5.9 s, +19 %) and loses more clock when packed (−21 % vs −6 %); its 384 threads deliver about what Graviton5's 192 single-threaded cores do (22.2 vs 22.9 jobs/s).

3.4. Density

- **Turin:** bursts of 3,000 (42.7 s) and 4,000 (81.9 s) with 0 failures; waves of 2,000 kept alive to **8,000 with 0 failures** (each wave ready in 10–15 s); peak **9,750** alive with 1.7 TB free. Configured limits found on the way: the API's 60 s create-pickup cap, the scratch reservation (1 GiB per sandbox → 6,742), `RLP_MAX_QUEUED_CREATES` = 8,000. The final stop was the runner's heartbeat starving under load: the API marked the runner lost and its lost-sweep marked 9,876 live sandboxes terminal, orphaning them (evidence added to 2459).
- **Vera / 9575F:** 2,000 alive, ≤ 1 failure per 16,000 jobs, ≈ 170–184 MiB host memory per sandbox under load. Configured cap 3,000.
- **Graviton5:** 2,000 with 0 failures in the startup test and the ladder (peak 504 of 754 GiB in the ladder). 64 KiB host pages cost ≈ 8 % more memory per sandbox.
- **Not measured:** idle-density ceiling on Vera and Graviton5.

4. Where the create time goes

From `POST /vms` to the first command answered, per sandbox; client timestamps joined with the runner's per-sandbox log on each host (same clock). Full tables in the appendix.

1,000 at once, idle image, fixes on (ms, p50)	HQ Vera	EPYC 9575F	Graviton5	Turin
A · API accepts the create (overlaps B)	285	652	196	252
B · request → runner starts the job (queue for a slot)	715	411	321	865
D2 · scratch + init-disk copy	11	38	14	10
D3 · Firecracker spawn + API socket	**208**	83	83	26
D4 · snapshot load + resume	32	39	16	**484**
D6 · cgroup, register, entrypoint trigger	124	155	78	17
D · runner clone total	441	413	246	706
E · running event → client sees it (SDK polling)	319	220	182	346
Total, request → first command done	**1,591**	**1,076**	**889**	**2,353**

- **Idle host, one sandbox:** ≈ 88 ms on Vera (API ≈ 5 ms, runner ≈ 60 ms with Firecracker spawn 39 ms, event back 2 ms, first command 21 ms). The harness reports ≈ 275 ms everywhere because its SDK first polls after 250 ms.
- **In a burst the time is waiting:** for a runner slot (B; 256 in flight by default) and for the next SDK poll (E).
- **Host-specific slow steps:** Vera's Firecracker spawn (208 ms, 2.5–8× the others) and Turin's snapshot load (484 ms vs 16–39 ms) — both unexplained and the obvious next investigations.

5. Fixes that changed the results

5.1. The four create-path fixes (each behind a flag; measurement builds, not merged)

Fix	What it changes	Effect
3260 `RLP_INITDISK_MODE=nofsync`	No fsync of the 16 MiB per-sandbox init disk (16 GB written per 1k burst)	D2: 352 ms (Vera), 567 ms (Graviton5), 1,293 ms (9575F) → 10–38 ms
3261 `RLP_CGROUP_POOL=1400`	Per-sandbox cgroups pre-created at runner start	188–201 ms → 24 ms on Vera
3265 `RLP_NEIGH_PERMANENT=1`	Permanent neighbour entry: removes a 580 ms post-resume stall on arm64	Vera single-create p99 658 → 73 ms
3269 `RLP_API_LISTEN_BACKLOG=4096`	API listen queue 128 → 4,096	Dropped connections 424–687 → 0 per burst

Together: 1k idle startup p50 −30 % on Vera, −57 % on the 9575F, ≈ −60 % on Graviton5, −36 % on Turin (all ready 7.95 → 3.88 s). No change to agent-image startup or throughput. On Turin only 3260 and 3261 mattered; 3265 and 3269 fix symptoms Turin never had.

5.2. Other fixes behind earlier result changes

Fix	Effect
Sudo storm (1754, upstream)	BM pair at 1,800 sandboxes: 1,720 of 1,880 failed → 0 of 3,600; p50 260 → 20 s
Firecracker socket wait 3 → 30 s (3212)	Spawn timeouts and runner self-cordons → 0 (Vera 177–304 per 1k); medians unchanged
Balloon off on 64 KiB-page hosts (1066)	Guest memory corruption 14/64 → 0/544
EROFS 4 KiB blocks (1060)	Vera creates: always failing → working
Busy loop `rlp-cpu-weight-floor.sh` stopped (9575F)	1k median 18.6 → 9.7 s; it ran during the published 2 Oct runs
ARP `gc_thresh3` 1,024 → 65,536	704 sandboxes: 1,324 failures → 0; peak 55.9 → 62.7 jobs/s
Memory burst cap 32 GiB → 512 MiB	2,112 sandboxes: 14.4 → 137.6 jobs/s; OOM kills 406 → 0
E2B huge pages released (HQ Vera)	Usable memory 2 GB → 1.44 TB
Hostname in `/etc/hosts`	`sudo` 113 s → 0.2 s; runner could not start without it
Client: 16 processes, open-files limit, SDK disk rounding	"Accept" 2.9–5.8 → 0.12–0.5 s; EMFILE failures → 0; Turin's density wall at 6,742 removed

6. Methodology corrections

- **The agent image measures its own workload, not startup.** Each sandbox burns 3–4 core-seconds at boot; with an idle image the platform starts 1,000 sandboxes in ≈ 4 s (fixes off) on Vera, the 9575F and Graviton5. Daytona's own CPU cost is ≈ 28 ms per sandbox.
- **"Control plane accepts the request" was the single-process Python client**, not the API.
- **Pinning to one socket helps only Vera, and only with CPU-heavy guests** (Vera's cross-socket memory costs 3.08× vs 1.25× on Graviton5); it halves throughput everywhere. The published 13.3 jobs/s was one Vera socket.

- **Published-data corrections:** the 9575F 2k mean kept a 151 s outlier and was taken with the busy loop on; the 9755′s 88.4 s drops a run (all four: 91.1 s); Vera's published 2k throughput point had 53–627 failures.
- **Measurement traps:** a second operator on HQ Vera invalidated one night of runs (rule: one operator per box); a 120 s client timeout reported healthy 9575F jobs as failures; SDK polling adds ≈ 0.2–0.35 s to every startup number.
- **Turin's "8 s startup" is explained:** it was measured without the fixes; with them it is 3.9 s.

7. Host page size (64 KiB vs 4 KiB)

Guests always use 4 KiB pages; the host page size only affects how the host backs and maps guest memory. On Graviton5 (same chip, same kernel version, one reboot) 64 KiB pages changed throughput by +1.5 % (noise), startup by +0.5 s and single-job time not at all, and cost ≈ 8 % more host memory per sandbox. The benchmark workload is CPU-bound with a small memory footprint, so cheaper address translation saves little. **On Vera itself this was never measured** — its 4 KiB kernel is installed but unused — so a gain there is unlikely for this workload but not ruled out.

8. What the data supports publicly

Claim	Verdict	Basis
Vera delivers high sustained throughput for CPU-intensive agent work	**supported**	Highest jobs/s of all hosts, flat from 352 to 2,000 sandboxes.
Vera outperforms leading x86 and Arm server CPUs on this work	**supported**, with scope	+20 % vs dual-socket Turin, +12–16 % vs Graviton5, 3.3× the 9575F — on Daytona's agent sandbox benchmark only.
Fast per-core performance	**supported**	+21–31 % jobs/s per physical core; fastest single job.
Sustains that speed with thousands of tasks running	**supported** as aggregate throughput	Throughput flat to 2,000; individual jobs get slower as more share the CPU (34 s per job at 2,000), as on every host.
Faster sandbox startup	**not supported**	Graviton5 and the 9575F start sandboxes faster; Turin under CPU load.
"Instantly", "without interruption", "without trading responsiveness for scale"	**not supported**	Absolute claims; Vera had rare guest wedges (3213) and per-job latency rises with load.
High-bandwidth memory makes it well suited to sandboxes	unproven	True as a spec; our CPU-bound workload did not exercise memory bandwidth.

9. Open questions and next tests

Test	Why	Time
Vera full ladder (44 ⟶ 2,000) + Vera c = 44 on the current build	Vera ran only 176 / 352 / 704 / 2,000; its single-job time is from the September build.	≈ 1 h
Clock under load, measured one way on every host	The −3 % / −6 % / −21 % / −24 % figures come from different methods.	with the ladder
Jobs per watt (Vera vs Turin vs 9575F)	NVIDIA's efficiency claim; Vera exposes per-socket power, AMD package power. Sampler ready (`gapfill-sampler.py`).	with the ladders
Memory-bandwidth-heavy sandbox workload	Vera's design point; the agent workload is ≈ 11 % memory-bound, so it cannot show it.	≈ 1 day
Equalize `RLP_VM_MAXACK` (2,048 on Vera)	The 9575F's startup lead may partly be its larger in-flight cap.	≈ 15 min
Vera 4 KiB vs 64 KiB host pages	Settles whether the large-page kernel helps Vera at all.	≈ 1.5 h
Vera Firecracker spawn / Turin snapshot load	The two unexplained slow create steps.	≈ 0.5 day each
3213 guest wedges, 2459 runner heartbeat at high density	Reliability on Vera; density ceiling above ≈ 8,000.	platform work

10. Appendix — create path, per step (ms, p50 / p90)

Steps: **A** API accepts the create (overlaps B) · **B** request sent ⟶ runner starts the job · **C** admission, image, template · **D1** netns · **D2** scratch + init disk · **D3** Firecracker spawn · **D4** snapshot load + resume · **D5** guest toolbox accepts · **D6** cgroup, register, trigger · **E** running event ⟶ client (SDK polling) · **F** first command. Medians of steps do not add up exactly to the median total. Turin: fixes-on only (its fixes-off runner log rotated out).

10.1. 10 sandboxes at once (low load, harness warm-up runs)

Step (ms, p50 / p90)	HQ Vera off	HQ Vera on	EPYC 9575F off	EPYC 9575F on	Graviton5 off	Graviton5 on
A · API accepts the create (POST answered)†	16 / 19	17 / 19	17 / 20	17 / 19	19 / 48	25 / 32
B · request sent ⟶ runner starts the job (API + JetStream queue)	7 / 8	6 / 7	7 / 9	7 / 8	7 / 16	12 / 13
C · runner admission, image + template lookup	1 / 1	0 / 1	1 / 1	0 / 1	1 / 1	1 / 1
D1 · network namespace from the pool	0 / 0	0 / 0	0 / 2	0 / 0	0 / 1	0 / 0
D2 · scratch copy + 16 MiB init disk (fsync)	31 / 31	5 / 5	38 / 41	7 / 8	30 / 32	10 / 13
D3 · Firecracker spawn + API socket	84 / 89	53 / 56	7 / 17	18 / 20	5 / 28	27 / 28
D4 · snapshot load + resume	4 / 6	4 / 5	6 / 8	9 / 11	12 / 58	6 / 7
D5 · guest toolbox port accepts	2 / 3	3 / 3	16 / 17	16 / 17	2 / 22	2 / 2
D6 · cgroup create/join, register, entrypoint trigger	8 / 35	3 / 9	13 / 15	3 / 5	21 / 24	2 / 21
D · runner clone total	131 / 135	68 / 70	81 / 83	52 / 58	74 / 119	48 / 52
E · running event ⟶ client sees it (NATS ⟶ API ⟶ SDK poll)	181 / 193	196 / 199	182 / 191	211 / 216	193 / 215	221 / 229
F · first command through the proxy (`echo ready`)	2 / 3	2 / 3	2 / 3	2 / 3	3 / 13	2 / 3
Total: request ⟶ first command done	273 / 274	274 / 275	273 / 275	273 / 274	276 / 318	279 / 294
samples	20	20	20	20	40	20

10.2. 1,000 at once, idle image

Step (ms, p50 / p90)	HQ Vera off	HQ Vera on	EPYC 9575F off	EPYC 9575F on	Graviton5 off	Graviton5 on	Turin on
A · API accepts the create (POST answered)†	324 / 1,208	285 / 438	349 / 1,169	652 / 774	143 / 208	196 / 334	252 / 340
B · request sent → runner starts the job (API + JetStream queue)	1,271 / 2,244	715 / 1,493	268 / 1,894	411 / 760	756 / 1,855	321 / 748	865 / 2,009
C · runner admission, image + template lookup	1 / 1	1 / 2	1 / 2	1 / 12	1 / 1	1 / 2	1 / 1
D1 · network namespace from the pool	0 / 0	0 / 0	0 / 0	0 / 6	0 / 0	0 / 0	0 / 0
D2 · scratch copy + 16 MiB init disk (fsync)	352 / 665	11 / 33	1,293 / 1,605	38 / 79	567 / 602	14 / 36	10 / 56
D3 · Firecracker spawn + API socket	56 / 143	208 / 320	10 / 27	83 / 193	5 / 16	83 / 136	26 / 105
D4 · snapshot load + resume	11 / 51	32 / 110	8 / 23	39 / 172	5 / 17	16 / 66	484 / 1,196
D5 · guest toolbox port accepts	3 / 9	6 / 510	17 / 19	18 / 30	4 / 22	28 / 43	16 / 19
D6 · cgroup create/join, register, entrypoint trigger	151 / 370	124 / 273	12 / 75	155 / 339	3 / 28	78 / 162	17 / 100
D · runner clone total	629 / 1,129	441 / 822	1,341 / 1,720	413 / 640	593 / 675	246 / 349	706 / 1,277
E · running event → client sees it (NATS → API → SDK poll)	421 / 805	319 / 686	474 / 909	220 / 408	282 / 1,026	182 / 476	346 / 1,025
F · first command through the proxy (`echo ready`)	4 / 7	4 / 9	3 / 7	5 / 11	5 / 24	15 / 48	4 / 8
Total: request → first command done	2,362 / 3,693	1,591 / 2,553	2,522 / 3,835	1,076 / 1,559	2,170 / 3,475	889 / 1,526	2,353 / 3,734
samples	3,000	2,999	3,000	3,000	3,000	3,000	2,000

10.3. 1,000 at once, agent image

Step (ms, p50 / p90)	HQ Vera off	HQ Vera on	EPYC 9575F off	EPYC 9575F on	Graviton5 off	Graviton5 on	Turin on
A · API accepts the create (POST answered)†	283 / 455	322 / 506	381 / 446	899 / 1,118	124 / 182	228 / 381	258 / 368
B · request sent → runner starts the job (API + JetStream queue)	4,187 / 13,736	2,446 / 18,456	230 / 5,126	563 / 4,133	949 / 4,419	685 / 5,067	970 / 3,102
C · runner admission, image + template lookup	1 / 2	1 / 2	1 / 1	1 / 5	1 / 1	1 / 2	1 / 1
D1 · network namespace from the pool	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 1
D2 · scratch copy + 16 MiB init disk (fsync)	218 / 741	14 / 133	1,008 / 1,578	54 / 143	299 / 612	12 / 34	17 / 72
D3 · Firecracker spawn + API socket	1,335 / 4,931	1,677 / 11,007	468 / 3,526	1,799 / 5,026	44 / 1,170	491 / 2,421	74 / 200
D4 · snapshot load + resume	211 / 2,373	36 / 650	114 / 257	130 / 301	30 / 606	30 / 476	866 / 1,981
D5 · guest toolbox port accepts	11 / 512	5 / 513	107 / 380	131 / 402	21 / 142	19 / 531	24 / 115
D6 · cgroup create/join, register, entrypoint trigger	1,121 / 4,832	372 / 2,929	2,976 / 8,759	1,736 / 7,021	565 / 2,726	561 / 2,499	49 / 161
D · runner clone total	5,232 / 9,350	4,318 / 12,153	6,237 / 11,677	5,487 / 11,296	1,655 / 3,952	2,050 / 4,066	1,188 / 2,204
E · running event → client sees it (NATS → API → SDK poll)	649 / 1,772	859 / 1,523	765 / 1,700	625 / 1,717	556 / 1,705	479 / 1,425	391 / 1,328
F · first command through the proxy (`echo ready`)	9 / 37	11 / 41	51 / 228	54 / 236	54 / 296	38 / 252	13 / 70
Total: request → first command done	11,846 / 21,834	11,725 / 23,940	7,949 / 17,934	6,765 / 16,566	3,558 / 9,469	5,506 / 9,626	2,556 / 5,767
samples	2,998	2,998	3,000	3,000	3,000	3,000	2,000

10.4. 2,000 at once, fixes on (idle: first three columns; agent: last three)

Step (ms, p50 / p90)	HQ Vera on	EPYC 9575F on	Turin on	HQ Vera on	EPYC 9575F on	Turin on
A · API accepts the create (POST answered)†	727 / 1,068	1,257 / 1,603	652 / 1,079	952 / 1,809	2,398 / 2,942	793 / 1,327
B · request sent → runner starts the job (API + JetStream queue)	1,858 / 4,106	868 / 2,357	2,284 / 4,565	15,616 / 40,159	5,957 / 25,924	3,946 / 11,721
C · runner admission, image + template lookup	1 / 1	1 / 2	0 / 1	1 / 3	1 / 1	1 / 1
D1 · network namespace from the pool	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 11
D2 · scratch copy + 16 MiB init disk (fsync)	16 / 63	21 / 61	10 / 36	29 / 761	72 / 194	42 / 98
D3 · Firecracker spawn + API socket	188 / 424	82 / 198	28 / 67	3,625 / 8,509	3,052 / 9,262	296 / 1,885
D4 · snapshot load + resume	37 / 335	152 / 442	657 / 1,248	140 / 1,791	430 / 1,064	255 / 1,912
D5 · guest toolbox port accepts	11 / 523	32 / 517	17 / 20	11 / 550	271 / 940	82 / 341
D6 · cgroup create/join, register, entrypoint trigger	104 / 284	85 / 330	9 / 77	694 / 2,855	5,984 / 16,766	249 / 2,101
D · runner clone total	589 / 1,098	700 / 1,244	789 / 1,316	6,285 / 11,616	11,483 / 23,433	2,189 / 4,166
E · running event → client sees it (NATS → API → SDK poll)	413 / 1,415	354 / 849	509 / 1,281	930 / 1,795	868 / 1,800	692 / 1,685
F · first command through the proxy (`echo ready`)	4 / 9	6 / 12	5 / 11	10 / 39	125 / 642	29 / 148
Total: request → first command done	2,965 / 6,315	2,403 / 3,486	3,496 / 6,754	27,620 / 46,069	19,879 / 50,431	7,073 / 17,184
samples	4,000	6,000	4,000	3,998	4,000	2,000

Sources: `vera-hq-cell-20261014/` (BASELINE-VERA/EPYC9575-FIXES, CREATE-PATH-STEPS, FINDINGS), `phoenix-e6-cell-20261014/FINDINGS.md`, `graviton5-cell-20261014/FINDINGS.md`, `gapfill-20261014/create-path-breakdown/` (runner journals, `analyze.py`, `agg.json`), `daniel-focus-here/public/` (published September data). Companion PDFs in `gapfill-20261014/`: comparison report v2, fixes and findings, create-path breakdown.